Test 12:

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

Caral: an ancient South American city

Huge earth and rock mounds rise out of the desert of the Supe Valley near the coast of Peru in South America. These immense mounds appear simply to be part of the geographical landscape in this arid region squeezed between the Pacific Ocean and the Andes mountains. But looks deceive. These are actually human-made pyramids. Strong evidence indicates they are the remains of a city known as Caral that flourished nearly 5,000 years ago. If true, it would be the oldest known urban center in the Americas and among the most ancient in the world.

Research undertaken by Peruvian archaeologist Ruth Shady suggests that the 150-acre complex of pyramids, plazas and residential buildings was a thriving metropolis when Egypt's great pyramids were still being built. Though discovered in 1905, for years Caral attracted little attention, largely because archaeologists believed the structures were fairly recent. But the monumental scale of the pyramids had long interested Shady, who began excavations at the site in 1996, about 22 kilometers from the coast and 190 kilometers north of Peru's capital city of Lima.

Shady and her crew searched for broken remains of the pots and containers that most such sites contain. Not finding any only made her more excited; it meant Caral could be what archaeologists term pre-ceramic, that is, existing before the advent in the area of pot-firing techniques. Shady's team undertook the task of excavating Piramide Mayor, the largest of the pyramids. After carefully clearing away many hundreds of years' worth of rubble and sand, they identified staircases, walls covered with remnants of colored woven bags and brickwork. In the foundations, they found the remains of grass-like reeds woven into bags. The original workers, she surmised, must have filled these bags with stones from a nearby quarry and laid

them atop one another inside retaining walls, gradually giving rise to the pyramid's immense structure.

Shady had samples of the reeds subjected to radiocarbon dating and found that the reeds were 4,600 years old. This evidence indicated that Caral was, in fact, more than 1,000 years older than what had previously been thought to be the oldest urban center in the Americas.

What amazed archaeologists was not just the age, but the complexity and scope of Caral. Piramide Mayor alone covers an area nearly the size of four football fields and is 18 meters tall. A nine-meter-wide staircase rises from a circular plaza at the foot of the pyramid, passing over three terraced levels until it reaches the top. Thousands of manual laborers would have been needed to build such a project, not counting the many architects, craftsmen, and managers.

Shady's team found the remains of a large amphitheater, containing almost 70 musical instruments made of bird and deer bones. Clearly, music played an important role in Caral's society. Around the perimeter of Caral are a series of smaller mounds and various buildings. These indicate a hierarchy of living arrangements: large, well-kept rooms atop pyramids for the elite, ground-level quarters for craftsmen, and shabbier outlying dwellings for workers.

But why had Caral been built in the first place? Her excavations convinced Shady that Caral once served as a trade center for the region, which extends from the rainforests of the Amazon to the high forests of the Andes. Shady found evidence of a rich trading environment, including seeds of the cocoa bush and necklaces of shells, neither of which was native to the immediate Caral area.

This environment gave rise to people who did not take part in the production of food, allowing them to become priests and planners, builders and designers. Thus occupational specialization, elemental to an urban society, emerged.

But what sustained such a trading center and drew travelers to it? Was it food? Shady and her team found the bones of small edible fish, which must have come from the Pacific coast to the west, in the excavations. But they also found evidence of squash, sweet potatoes and beans having been grown locally. Shady theorized

that Caral's early farmers diverted the area's rivers into canals, which still cross the Supe Valley today, to irrigate their fields.

But because she found no traces of maize, which can be traded or stored and used in times of crop failure, she concluded that Caral's trade leverage was not based on stockpiling food supplies.

It was evidence of another crop in the excavations that gave Shady the best clue to Caral's success. In nearly every excavated building, her team discovered evidence of cotton — seeds, fibers and textiles. Her theory fell into place when a large fishing net made of those fibers, unearthed in an unrelated dig on Peru's coast, turned out to be as old as Caral.

"The farmers of Caral grew the cotton that the fishermen needed to make their nets," Shady speculates. "And the fishermen gave them shellfish and dried fish in exchange for these nets."

In essence, the people of Caral enabled fishermen to work with larger and more effective nets, which made the resources of the sea more readily available, and the fishermen probably used dried squash grown by the Caral people as flotation devices for their nets.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–6 on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1. Caral was built at the same time as the construction of the Egyptian pyramids.

2. The absence of pottery at the archaeological dig gave Shady a significant clue to the age of the site.

3. The stones used to build Piramide Mayor came from a location far away.

4. The huge and complicated structures of Piramide Mayor suggest that its construction required an organised team of builders.

5. Archaeological evidence shows that the residents of Caral were highly skilled musicians.

6. The remains of housing areas at Caral suggest that there were no class distinctions in residential areas.

Questions 7–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 7–13 on your answer sheet.

Caral as a trading centre

Items discovered at Caral but not naturally occurring in the area

- the **7.** _____ of a certain plant
- **8.** _____ used to make jewellery
- the remains of certain food such as **9.** _____

Clues to farming around Caral

- **10.** _____ still in existence today indicate water diverted from rivers
- no evidence that **11.** _____ was grown

Evidence of relationship with fishing communities

- the excavation findings and fishing nets found on the coast suggest Caral farmers traded **12.** _____
- dried squash may have been used to aid **13.** _____ of fishing nets

ĐÁP ÁN:

1. TRUE (Đoạn 2: "...was a thriving metropolis when Egypt's great pyramids were still being built.")

2. TRUE (Đoạn 3: "Not finding any (pots) only made her more excited; it meant Caral could be... pre-ceramic.")

3. FALSE (Đoạn 3: "...filled these bags with stones from a **nearby** quarry.")

4. TRUE (Đoạn 4: "Thousands of manual laborers would have been needed... not counting the many architects, craftsmen, and managers.")

5. NOT GIVEN (Đoạn 4 có nhắc đến âm nhạc quan trọng nhưng không khẳng định cư dân là những nhạc sĩ "có tay nghề cao" — highly skilled.)

6. FALSE (Đoạn 4: "...indicate a **hierarchy** of living arrangements" — có sự phân chia giai cấp giữa tầng lớp thượng lưu, thợ thủ công và công nhân.)

7. seeds (Đoạn 5: "...including seeds of the cocoa bush...")

8. shells (Đoạn 5: "...and necklaces of shells...")

9. fish (Đoạn 6: "...found the bones of small edible fish, which must have come from the Pacific coast...")

10. canals (Đoạn 6: "...diverted the area's rivers into canals, which still cross the Supe Valley today...")

11. maize (Đoạn 6: "...because she found no traces of maize...")

12. cotton (Đoạn 7: "The farmers of Caral grew the cotton that the fishermen needed to make their nets.")

13. flotation (Đoạn 7: "...fishermen probably used dried squash... as flotation devices for their nets.")

READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

Roller coaster: the great fairground attraction

How they move

Like a passenger train, a roller coaster consists of a series of connected cars that move on a track. But unlike a passenger train, it has no engine or power source of its own. For most of the ride, it is moved only by the forces of inertia and gravity. The only exertion of energy occurs at the very beginning of the ride when the coaster train is pulled up the lift hill.

The traditional lifting mechanism is a long length of chain running up the hill under the track. The chain is fastened in a loop, which is wound around a gear at the top of the hill and another one at the bottom. The gear at the bottom of the hill is turned by a motor. This turns the chain so that it continually moves up the hill like a long conveyor belt. The coaster cars grip onto the chain, which simply pulls them to the top. At the summit, the train is released and starts its descent.

The purpose of this initial ascent is to build up a sort of reservoir of potential energy, which simply means that as the coaster gets higher in the air, there is a greater distance gravity can pull it down.

As the train starts coasting down the hill, this potential energy is converted into kinetic energy (energy of motion), and the train speeds up. At the bottom of the hill, this has reached its maximum, and this propels the train up the second hill, again building up the potential-energy level.

In this way, the course of the track is constantly converting energy from kinetic to potential and back again. This fluctuation in acceleration is what makes roller coasters so much fun. At its most basic level, this is all a roller coaster is – a machine that uses gravity and inertia to send a train along a winding track.

Coasting through history

Roller coasters have a long, fascinating history. Their direct ancestors were ice slides, popular in Russia in the 16th and 17th centuries. They consisted of a long, steep, wooden slide covered in ice. Riders walked up a ladder or set of stairs to the top of the slide, as high as 21 metres up. At the top, they climbed into sleds made of wood or blocks of ice and shot down the slope. At the base, the sleds would crash-land in a sand pile.

It seems that the idea was then imported into France. For most of the year, the warmer climate would melt the ice, so the French started building waxed slides instead. To help the sleds move down these slides, they added wheels, and in 1817, for the first time, a train was attached to the track. The French continued to expand on this idea, coming up with more complex track layouts, with multiple cars and all sorts of twists and turns.

The first American roller coaster was built in the mountains of Pennsylvania in the mid-1800s, originally to provide an easy way to send coal to the railway 29 km down the mountain. When the track was first built, a crew at the bottom of the mountain would attach the cart to a team of mules after emptying the load, and the mules would drag it back up to the top. They were eventually replaced with steam engines to make the system more efficient.

Soon after these improvements were made, the railway company built a new tunnel that brought the freight trains much closer to the coal mine. No longer required for its original purpose, the roller coaster was configured as a ‘scenic tour’. For one dollar, tourists got a leisurely ride up to the top of the mountain, followed by a wild, bumpy ride straight down. This was soon a resounding success, attracting thousands of tourists every year.

Scenic rides like this continued to thrive and were joined by wooden roller coasters similar to the ones we know today. These coasters were the main attraction at popular amusement parks throughout the United States, such as the many parks of Coney Island in New York. By the 1920s, roller coasters were in full swing, with some 2,000 rides in operation around the country.

Following the Great Depression, a decline in roller-coaster production began in the early 1930s, but a second roller-coaster boom in the 1970s and the beginning of the 1980s revitalised the amusement-park industry and introduced a slew of innovative tubular steel coasters.

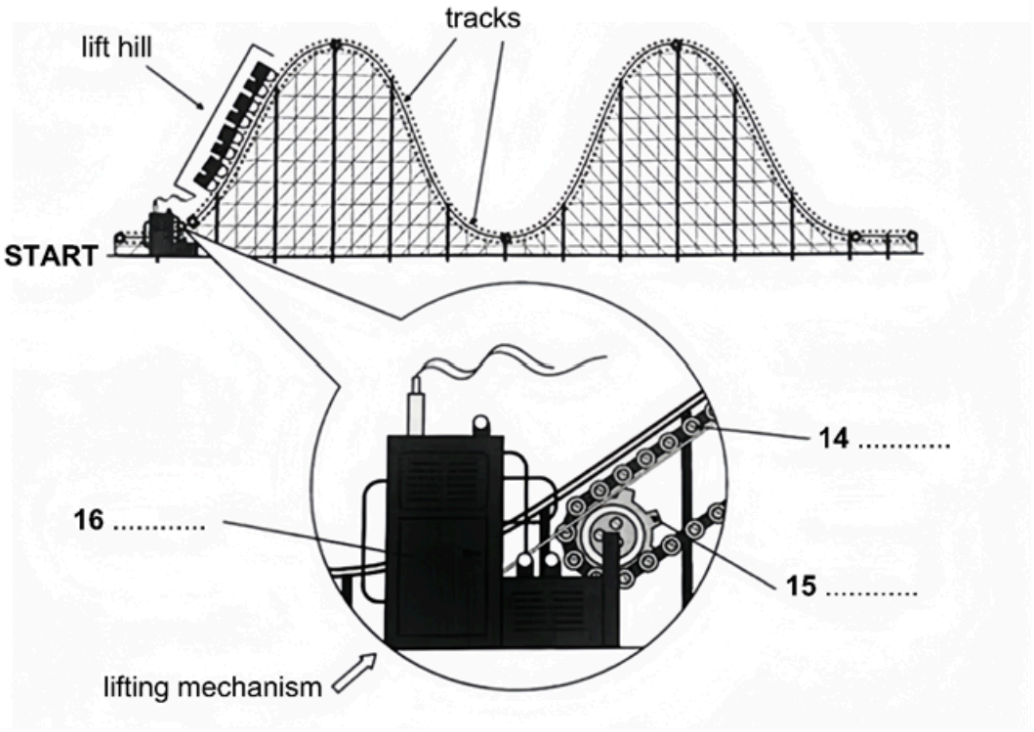
This was followed by a decline in interest for the rest of the decade, but since the early 1990s the amusement-park industry has experienced another coaster boom of sorts. New launching techniques and other technological developments have opened up a world of options for designers, so in some rides you feel as if you are flying. In the next few years we can expect to see many faster, taller and more twisted rides popping up in amusement parks around the world.

Questions 14–16

Label the diagram below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 14–16 on your answer sheet.



Questions 17–21

Complete the summary below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

Write your answers in boxes 17–21 on your answer sheet.

History of roller coasters

- Modern roller coasters are descended from 16th-century Russian slides with a surface of **17** _____.
- In France, because of the higher temperatures, the wooden surface on the slides was **18** _____, and **19** _____ were attached to the cars to ease the descent.
- The first US roller coaster was used for transporting **20** _____ down a mountainside in carts.
- Initially, these were pulled by mules, but in time power was produced by **21** _____.

Questions 22–26

Do the following statements agree with the information given in Reading Passage 2?

In boxes 22–26 on your answer sheet, write

TRUE *if the statement agrees with the information*
FALSE *if the statement contradicts the information*
NOT GIVEN *if there is no information on this*

22 The earliest modifications to the basic slide were made in France.

23 Roller coasters continued to increase in popularity throughout the 1920s and 1930s.

24 New roller-coaster technology was introduced in the 1970s in response to public demand.

25 Roller coasters were less popular for most of the 1980s than in the 1970s and 1990s.

26 The design of roller-coaster rides became more varied in the 1990s.

TRUE IELTS

ĐÁP ÁN:

14. chain: Sợi dây xích dài chạy dưới đường ray để kéo đoàn tàu lên

Đoạn chứa đáp án:

The traditional lifting mechanism is a long length of chain running up the hill under the track

15. gear: Bánh răng ở phía dưới (và phía trên) để cuốn sợi xích

Đoạn chứa đáp án:

The chain is fastened in a loop, which is wound around a gear at the top of the hill and another one at the bottom of the hill

16. motor: Động cơ dùng để làm quay bánh răng ở phía dưới .

Đoạn chứa đáp án:

The gear at the bottom of the hill is turned by a motor

17. ice: Tàu lượn hiện đại bắt nguồn từ các máng trượt băng ở Nga vào thế kỷ 16

Đoạn chứa đáp án:

They consisted of a long, steep, wooden slide covered in ice

18. waxed: Ở Pháp, do thời tiết ấm hơn làm tan băng, họ đã thay bằng các máng trượt **được bôi sáp**

Đoạn chứa đáp án:

For most of the year, the warmer climate would melt the ice, so the French started building waxed slides instead

19. wheels: Người Pháp đã gắn thêm **bánh xe** vào các xe trượt để hỗ trợ việc đi xuống

Đoạn chứa đáp án:

To help the sleds move down these slides, they added wheels

20. coal: Tàu lượn đầu tiên tại Mỹ được dùng để vận chuyển **than** xuống núi

Đoạn chứa đáp án:

originally to provide an easy way to send coal to the railway

21. steam engines: Ban đầu xe được kéo bởi lừa, nhưng sau đó được thay thế bằng **động cơ hơi nước**

Đoạn chứa đáp án:

They were eventually replaced with steam engines

22. TRUE: Những sửa đổi sớm nhất đối với máng trượt cơ bản (như thêm bánh xe, bôi sáp) được thực hiện tại Pháp

Đoạn chứa đáp án:

It seems that the idea was then imported into France. For most of the year, the warmer climate would melt the ice, so the French started building waxed slides instead.

23. FALSE: Tàu lượn không duy trì sự gia tăng phổ biến trong suốt những năm 1930; một sự sụt giảm đã bắt đầu từ đầu thập kỷ này sau cuộc Đại khủng hoảng

Đoạn chứa đáp án:

By the 1920s, roller coasters were in full swing, with some 2,000 rides in operation around the country. Following the Great Depression, a decline in roller coaster production began in the early 1930s

24. NOT GIVEN: Bài đọc có nhắc đến công nghệ tàu lượn thép mới trong những năm 1970 nhưng không nói rõ liệu nó có phải được ra đời để "đáp ứng nhu cầu của công chúng" hay không

Đoạn tham khảo:

a second roller-coaster boom in the 1970s and the beginning of the 1980s revitalized the amusement park industry, and introduced a slew of innovative tubular steel coasters. (public demand)

25. TRUE: Tàu lượn ít phổ biến hơn trong phần lớn những năm 1980 (thời kỳ sụt giảm sự quan tâm) so với những năm 1970 và 1990 (hai thời kỳ bùng nổ)

Đoạn chứa đáp án:

This was followed by a decline in interest for the rest of the decade, but since the early 1990s the amusement-park industry has experienced another coaster boom of sorts.

26. TRUE: Thiết kế tàu lượn trở nên đa dạng hơn vào những năm 1990 nhờ các kỹ thuật phóng mới và phát triển công nghệ

Đoạn chứa đáp án:

New launching techniques and other technological developments have opened up a world of options for designers so in some rides you feel as if you are flying.

READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

Risk taking

A

We run risks. Most of the time we ignore them; sometimes we stop and decide how to deal with them. That's life. Risks arise in sports, in business, in personal relationships; they are present in the laboratory, in political calculations, everywhere. Acting individually, we often prefer to gain some benefit by accepting some danger. Acting collectively, we make similar trade-offs.

B

When society must assess risks, and those risks can be clarified, at least in part, by science, technology or mathematics, the responsibility of the research community is to be clear and complete. But that is not always so easy. Furthermore, too few people pay full attention to the best evidence, much less modify their behaviors as a result. Even when the evidence of high risk is clear – about smoking, for instance – many will deny the risk, or dispute it, or accept that it exists and still go ahead. As a former cigarette smoker, I understand denial.

C

Perhaps more important than the basic facts of risk are the varied perception of it. Hearing the prediction of 60% chance of rain, or recognizing the minuscule likelihood of becoming ill from a flu shot, most people understand the right responses: take your umbrella and get your flu shot. But some, fearfully but irrationally, go to the extreme and stay at home and avoid all contact with other people. Irrational decisions occasionally affect us all. When it comes to risks the best odds do not always influence people to make the correct decision.

D

In her book, *The Universe and the Teacup*, the science writer K. C. Cole reminds us that the risks of losing something and gaining something appear to be calculated in our minds according to quite different scales. People will risk a lot to prevent a loss, she notes; some people will rush into a burning building to save a pet. But they will usually risk very little if the only perceived outcome is a possible gain; many people won't even risk pursuing an exciting job, if the current job, though unfulfilling, remains comfortable. That psychological mechanism, a desire to maintain the status quo, helps explain the high hurdles that any rational risk assessment must confront.

E

Organizations face the same hurdles whenever they must cope with major shifts in policy. In his classic examination of the character of high-tech industry, *Only the Paranoid Survive*, Andrew S. Grove gives a perceptive account of a crisis that took place during his tenure as Chairman of the Intel Corporation. The business environment for Intel and its competitors

was transformed by a flood of low-cost memory chips from Japan, and Intel made the difficult decision to abandon its chip-making business. Such a period of change in the outside world is filled with danger, Grove notes, precisely because it is so easy for executives to try to disregard it or misinterpret the mounting evidence showing the profound risks of not making change at the very core. The distinction Cole cites is instructive here: one reason management and staff may not act is that they tend to focus on the potential gains of risking change (which they characterize as small), rather than on the catastrophic losses that risking change could prevent. The period becomes critical for a company, according to Grove, and deadly when unattended to.

F

Likewise, at the national level, and even the global one, institutions tend to be more willing to take a risk if it means avoiding a loss, than they are if it might confer a gain. Almost everyone, for instance, sees the signs of ecological decay. Reid J. Lifset's article, 'Full Accounting', describes newly developed techniques for clarifying which critical decision would have the highest pay-offs for the environment. Once these 'leverage points' as he calls them, are located, many environmental organizations will be willing to risk the resources needed to prevent further deterioration.

G

More complicated are decisions by nations to invest broadly in educational research. Some governmental leaders are oblivious to the risks posed by social under-investment, because the urgent needs of the present take precedence over actions that would give rise to even more substantial gains in the future. Their attitude is that today's risks must be manageable, and that tomorrow's gains are at the very least difficult to measure. Better consideration and decision making for what lies ahead is essential.

H

Nevertheless, I confess to being an optimist. My glass is generally half full – though that may not help me handle risk. Whether one's temperament is optimistic or pessimistic, everyone in our scientific and engineering community has a social responsibility: measure with care and portray with honesty the level of risk involved in any action or decision.

Questions 27–30

*Complete each sentence with the correct ending, **A–G**, below.*

*Write the correct letter, **A–G**, in boxes 27–30 on your answer sheet.*

27 K. C. Cole points out that avoiding a loss seems more important to people than

28 According to Andrew S. Grove, it can be very dangerous for companies to ignore

29 Reid J. Lifset argues that environmental agencies can benefit from

30 The writer suggests that government leaders need to focus more on

A planning for the future.
B predicting their errors.
C continuing in the present situation.
D recent findings on the most effective solutions to problems.
E achieving a gain.
F changes occurring externally.
G a high level of agreement.

TRUE IELTS

Questions 31–39

*Reading Passage 3 has eight paragraphs, **A–H**.*

Which paragraph contains the following information?

*Write the correct letter, **A–H**, in boxes 31–39 on your answer sheet.*

***NB** You may use any letter more than once.*

31 the areas of study that help us understand risk

32 reference to key actions that might benefit the world

33 examples of statistical predictions that the majority of the public react to in the same way

34 the variety of contexts in which risk occurs

35 mention of a habit that has been proved to be very risky

36 reference to market forces motivating change

37 the writer's current attitude towards risk

38 a reason why some individuals avoid making a career change

39 how some people over-react to information about risk

Question 40

*Choose the correct letter, **A, B, C or D**.*

Write the correct letter in box 40 on your answer sheet.

At the end of the passage the writer concludes that

A risk is something we must try to avoid.
B experts should be truthful about risk to the public.
C optimists are better able to deal with risk than pessimists.
D it is important to take risks in order to fulfil social obligations.

ĐÁP ÁN:

27. E (achieving a gain): K. C. Cole chỉ ra rằng việc tránh một tổn thất có vẻ quan trọng với con người hơn là **đạt được một lợi ích**

28. F (changes occurring externally): Theo Andrew S. Grove, sẽ rất nguy hiểm nếu các công ty phớt lờ **những thay đổi đang diễn ra bên ngoài**

29. D (recent findings on the most effective solutions to problems): Reid J. Lifset lập luận rằng các tổ chức môi trường có thể hưởng lợi từ **những phát hiện gần đây về các giải pháp hiệu quả nhất cho các vấn đề**

30. A (planning for the future): Tác giả gợi ý rằng các nhà lãnh đạo chính phủ cần tập trung nhiều hơn vào việc **lập kế hoạch cho tương lai**

31. B: Các lĩnh vực nghiên cứu giúp chúng ta hiểu về rủi ro (nhắc đến khoa học, công nghệ, toán học)

32. F: Đề cập đến các hành động then chốt có thể mang lại lợi ích cho thế giới (các "điểm đòn bẩy" cho môi trường)

33. C: Ví dụ về các dự báo thống kê mà đa số công chúng phản ứng theo cùng một cách (ví dụ về dự báo mưa 60% hoặc tiêm ngừa cúm)

34. A: Sự đa dạng của các bối cảnh mà rủi ro xảy ra (thể thao, kinh doanh, quan hệ cá nhân...)

35. B: Nhắc đến một thói quen đã được chứng minh là rất rủi ro (hút thuốc lá)

36. E: Đề cập đến các lực lượng thị trường thúc đẩy sự thay đổi (sự tràn ngập chip giá rẻ từ Nhật Bản)

37. H: Thái độ hiện tại của tác giả đối với rủi ro (thú nhận mình là người lạc quan)

38. D: Lý do tại sao một số cá nhân tránh thay đổi nghề nghiệp (mong muốn duy trì hiện trạng/sự thoải mái dù không thỏa mãn)

39. C: Cách một số người phản ứng thái quá với thông tin về rủi ro (cực đoan đến mức ở nhà và tránh tiếp xúc mọi người)

40. B: Ở cuối bài đọc, tác giả kết luận rằng **các chuyên gia nên thành thực về rủi ro đối với công chúng** (Đoạn H: "measure with care and portray with honesty the level of risk")

TRUE IELTS